# Multi-omics dissection of therapeutic resistance mechanisms in KRAS-mutant cancers to identify rational combination strategies

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**One sentence summary**

This proposal employs multi-omics profiling to systematically characterize mutation-specific resistance mechanisms in KRAS-mutant cancers and identify rational combination therapy strategies for patients who rapidly develop resistance to current G12C inhibitors or harbor untreatable non-G12C mutations

**Abstract**

KRAS-mutant cancers affect over 175,000 new cases annually in the United States, yet therapeutic options remain severely limited by rapid resistance development. Despite FDA approval of KRAS G12C inhibitors (sotorasib, adagrasib), median progression-free survival is only 3-6 months, with patients developing resistance through multiple adaptive mechanisms including EGFR-mediated MAPK reactivation, PI3K/AKT hyperactivation, lipid-dependent KRAS relocalization, and epigenetic reprogramming. Furthermore, 85% of KRAS-mutant patients harbor non-G12C mutations (G12D, G12V, G13D) for which no direct inhibitors exist, necessitating alternative therapeutic strategies.

This proposal addresses the fundamental challenge of therapeutic resistance in KRAS-driven cancers through systematic multi-omics profiling of resistance mechanisms. We hypothesize that distinct KRAS mutations generate mutation-specific resistance vulnerabilities that can be exploited through rational combination therapies. Using isogenic cell lines harboring G12C, G12D, G12V, and G13D mutations across pancreatic, colorectal, and lung cancer backgrounds, we will employ four complementary omics approaches to dissect resistance at multiple molecular layers.

Transcriptomics will identify compensatory pathway activation and adaptive transcriptional programs that enable therapeutic escape. Computational modeling suggests approximately 168 genes may be upregulated in KRAS mutants versus wild-type, including ERK-MAPK bypass targets and metabolic adaptation genes that sustain growth under targeted inhibition. Phosphoproteomics will map kinase signaling networks that reactivate oncogenic signaling despite KRAS inhibition. Lipidomics will characterize lipid remodeling that enables KRAS membrane relocalization and sustained nanoclustering under therapeutic pressure. Published studies suggest 2-3-fold elevation in glycosphingolipids (GM3) critical for maintaining KRAS plasma membrane localization, revealing vulnerabilities through FASN or glycosphingolipid synthesis inhibition to disrupt KRAS membrane anchoring during resistance. Epigenomics will reveal chromatin remodeling and epigenetic plasticity that stabilize resistance-associated transcriptional states. ATAC-seq and ChIP-seq will identify druggable epigenetic regulators whose inhibition could reverse or prevent resistance.

This integrated approach will deliver: (1) resistance prediction biomarker panels for early therapeutic intervention; (2) rational combination therapy strategies targeting adaptive resistance pathways; (3) mutation-specific therapeutic vulnerabilities for non-G12C patients; and (4) novel targets in lipid metabolism and epigenetic machinery that destabilize resistance phenotypes. Findings will have immediate applicability to ongoing clinical trials and address urgent needs for the majority of KRAS-mutant cancer patients who lack effective treatment options.

**Problem Statement**

The Kirsten rat sarcoma viral oncogene homolog (KRAS) encodes a membrane-bound GTPase that functions as a molecular switch in cellular signaling, alternating between an inactive GDP-bound state and an active GTP-bound state. Oncogenic mutations lock KRAS in its active conformation, driving constitutive signaling through the MAPK and PI3K/AKT pathways and promoting uncontrolled cell proliferation (Huang et al., 2021; Yang et al., 2023).

KRAS mutations represent a substantial clinical burden, occurring in approximately 23% of all adult cancers, with highest prevalence in pancreatic ductal adenocarcinoma (92%), colorectal cancer (49%), and non-small cell lung cancer (35%), accounting for over 175,000 new cases annually in the United States. (Lee et al., 2022). The most common mutations G12D (29%), G12V (23%), and G12C (15%) comprise 68% of all KRAS alterations and exhibit distinct biochemical properties: G12 mutations demonstrate impaired intrinsic GTPase activity, while G13D mutations retain partial sensitivity to GTPase activating protein (GAP) regulation. These biochemical differences translate into mutation-specific signaling profiles, with G12C and G12V showing increased RAL signaling and decreased AKT phosphorylation compared to G12D. (Lee et al., 2022; Yang et al., 2023; Huang et al., 2021).

Despite decades of research, KRAS remained "undruggable" until the recent development of covalent G12C inhibitors (sotorasib and adagrasib), approved in 2021-2022 (Yang et al., 2023). However, significant limitations persist: these inhibitors are mutation-specific, benefiting only patients with G12C mutations; median progression-free survival remains modest at 6 months in Non-Small Cell Lung Cancer (NSCLC) and 3 months in colorectal cancer; and acquired resistance develops rapidly through multiple mechanisms including EGFR-mediated MAPK pathway rescue, PI3K/AKT hyperactivation bypassing KRAS dependence, secondary KRAS mutations, and activation of alternative RTKs. Over 85% of KRAS-mutant patients harbor non-G12C mutations for which no direct inhibitors exist, and co-alterations in genes like STK11 and KEAP1 reduce response to both KRAS inhibitors and immunotherapy. (Wang et al., 2025; Yang et al., 2023).

Current understanding of resistance mechanisms remains incomplete at multiple molecular levels critical for designing rational combination therapies. While transcriptomic studies have identified mutation-specific gene signatures associated with differential drug response, the proteome-level consequences including post-translational modifications that enable kinase network rewiring and protein complex dynamics that sustain oncogenic signaling under inhibitor pressure remain poorly characterized. (Huang et al., 2021). Epigenetic alterations that stabilize resistance-associated transcriptional states and enable cells to adapt to therapeutic pressure remain largely unexplored, despite growing evidence that epigenetic plasticity drives resistance in other targeted therapies. Similarly, lipid-dependent regulation of KRAS membrane localization critical for nanocluster formation, signaling fidelity, and drug sensitivity remains understudied despite evidence that different KRAS mutants exhibit allele-specific lipid preferences that may create distinct vulnerabilities or resistance mechanisms. A comprehensive multi-omics characterization integrating transcriptomics, proteomics, metabolomics, and epigenomics is essential to generate a molecular atlas of KRAS-mutant cancers, identify mutation-specific dependencies, uncover mechanisms of drug resistance, and reveal novel therapeutic vulnerabilities beyond direct KRAS inhibition.

**Project Goals**

**Systematically Map Adaptive Resistance Mechanisms Through Multi-Omics Integration:** We will systematically compare drug-naive and resistance-evolved models across all omics layers. Transcriptomics will identify compensatory pathway activation bypassing KRAS inhibition. Phosphoproteomics will reveal kinase network rewiring reactivating oncogenic signaling despite KRAS blockade. Lipidomics will uncover metabolic rewiring altering KRAS membrane localization during resistance evolution. Epigenomics will map chromatin remodeling stabilizing resistance-associated gene expression. This integrated analysis will reveal temporal resistance dynamics, identify early predictive biomarkers, and nominate rational combination therapy targets.

**Exploit Lipid-Dependent Vulnerabilities to Disrupt KRAS Membrane Anchoring:** KRAS membrane localization and nanoclustering depend on specific lipid environments regulating signaling fidelity and drug sensitivity. Different mutants exhibit allele-specific lipid preferences, and cells may adapt lipid composition during resistance. Our lipidomic profiling will define mutation-specific and resistance-associated lipid dependencies governing KRAS membrane anchoring. We will measure glycosphingolipids, phosphatidylserine species, and cholesterol in plasma-membrane-enriched fractions comparing drug-naive versus resistant cells. Perturbation experiments will test whether inhibiting lipid synthesis disrupts KRAS localization and sensitizes resistant cells. Targeting the membrane environment required for KRAS function may be less susceptible to point-mutation-based resistance than direct KRAS inhibition.

**Identify Epigenetic Vulnerabilities That Reverse Resistance Through Chromatin Reprogramming:** Epigenetic plasticity enables cells to rapidly adapt transcriptional states under drug pressure without genetic mutations. Our epigenomic profiling using ATAC-seq and ChIP-seq will map chromatin accessibility and histone modification changes comparing drug-naive versus resistant states. We will identify resistance-associated enhancers and promoters becoming accessible during therapy adaptation. Motif enrichment will nominate transcription factors driving resistance programs. Integration with transcriptomic data will link epigenetic changes to functional resistance gene expression. Identifying druggable epigenetic regulators whose inhibition reverses resistance-associated chromatin states offers combination therapy opportunities.

**Address the Non-G12C Therapeutic Gap Through Mutation-Specific Vulnerabilities:** Current FDA-approved KRAS inhibitors exclusively target G12C, leaving 85% of patients with G12D, G12V, and G13D mutations without direct options. By profiling transcriptomic, proteomic, lipidomic, and epigenomic landscapes across KRAS subtypes, we will identify mutation-specific dependencies creating allele-selective therapeutic opportunities. Published studies indicate suggest G12D exhibits distinct PI3K/AKT activation versus G12C, while G12V shows preferential RAL pathway engagement. Understanding intrinsic differences between mutations reveals pre-existing vulnerabilities and predicts likely resistance mechanisms.

**Why fund this?** KRAS-mutant cancers represent 500,000 new cases globally annually with poor outcomes. Rapid resistance emergence threatens to undermine the decade-long effort to drug KRAS. This systematic approach will reveal rational combination strategies preventing resistance in G12C patients, identify actionable targets for non-G12C patients, and uncover lipid-dependent and epigenetic vulnerabilities as alternative therapeutic entry points less susceptible to mutation-based resistance, delivering actionable insights for improved therapeutic durability and patient outcomes.

**Methods:**

**Transcriptomics: Identifying Compensatory Pathways Driving Therapeutic Resistance**

A major resistance mechanism to KRAS inhibitors involves transcriptional reprogramming that activates compensatory signaling pathways, bypassing direct KRAS activity. To systematically identify these adaptive programs and nominate combination therapy targets blocking resistance emergence, we will perform bulk RNA sequencing comparing drug-naive and resistance-evolved models across KRAS mutations.

**Experimental Design and Methods:** We will perform bulk RNA-seq on isogenic cell line panels harboring G12C, G12D, G12V, and G13D mutations compared to WT controls, representing PDAC, CRC, and NSCLC backgrounds to capture tissue-specific effects. Total RNA will be isolated using TRIzol, quality-assessed via Bioanalyzer (RIN > 8), and prepared using Illumina TruSeq Stranded mRNA Kit. Sequencing will employ NovaSeq 6000 platform (150 bp paired-end; 30-40M reads/sample; n=3 replicates). Reads will be trimmed using Trimmomatic, aligned with STAR, and analyzed with DESeq2. Pathway enrichment via GSEA will highlight resistance-associated activation of MAPK/ERK bypass pathways, PI3K/AKT hyperactivation, and RAL-GEF signaling. Top candidate resistance genes will be validated by RT-qPCR and functionally tested through CRISPR knockout.

**Hypotheses:** 1) Resistance-evolved cells will show upregulation of alternative RTKs (EGFR, MET, HER2/3) reactivating MAPK signaling independently of KRAS; (2) PI3K/AKT pathway components will be transcriptionally elevated in resistant cells, particularly in G12D-mutant contexts; (3) Metabolic genes enabling adaptation to altered bioenergetic demands will be upregulated; (4) Mutation-specific resistance signatures will emerge, with G12V showing preferential RAL-GEF activation and G13D exhibiting anti-apoptotic gene expression.

**Impact:** Bulk RNA-seq comparing naive versus resistant states identifies transcriptional changes causally linked to resistance. These signatures serve as: (1) predictive biomarkers for early resistance detection; (2) rational combination therapy targets; and (3) mechanistic context for integrating proteome-level and epigenetic resistance mechanisms.

**Interpretation / Hypothesis Support:** These three transcriptomic visualizations collectively validate the central hypothesis that KRAS mutations drive compensatory transcriptional reprogramming that activates resistance pathways bypassing direct KRAS inhibition. The volcano plot Figure 1A identifies 328 significantly altered genes (162 upregulated, 166 downregulated) including potential RTK bypass candidates (EGFR, MET, HER2/3), PI3K/AKT pathway components, and metabolic adaptation genes the precise resistance mechanisms predicted in the proposal hypothesis. The PCA plot Figure 1B demonstrates the clear separation between KRAS A146 and control samples (PC1: 12.0%, PC2: 11.3% variance explained), confirming that oncogenic KRAS drives robust gene expression signatures independent of tissue context, consistent with mutation-driven transcriptional reprogramming. The heatmap Figure 1C reveals coordinated upregulation patterns across all KRAS replicates in the top 50 most significant genes, providing visual evidence of stable transcriptional programs consistent with MAPK/ERK bypass, PI3K/AKT hyperactivation, and RAL-GEF signaling the compensatory pathways hypothesized to enable therapeutic resistance.

Future Gene Set Enrichment Analysis (GSEA) will test whether these individual gene changes represent coordinated pathway-level activation by calculating enrichment scores for RTK signaling, PI3K/AKT/mTOR, and metabolic pathways, providing statistical validation that entire resistance mechanisms rather than isolated genes are operating in KRAS-mutant cells and nominating specific combination therapy strategies targeting multiple nodes within enriched pathways to prevent resistance emergence. Together, these findings nominate the 162 upregulated genes as rational combination therapy targets for RT-qPCR validation and CRISPR knockout experiments, directly addressing the 85% therapeutic gap for non-G12C KRAS-mutant patients who lack access to G12C-specific inhibitors

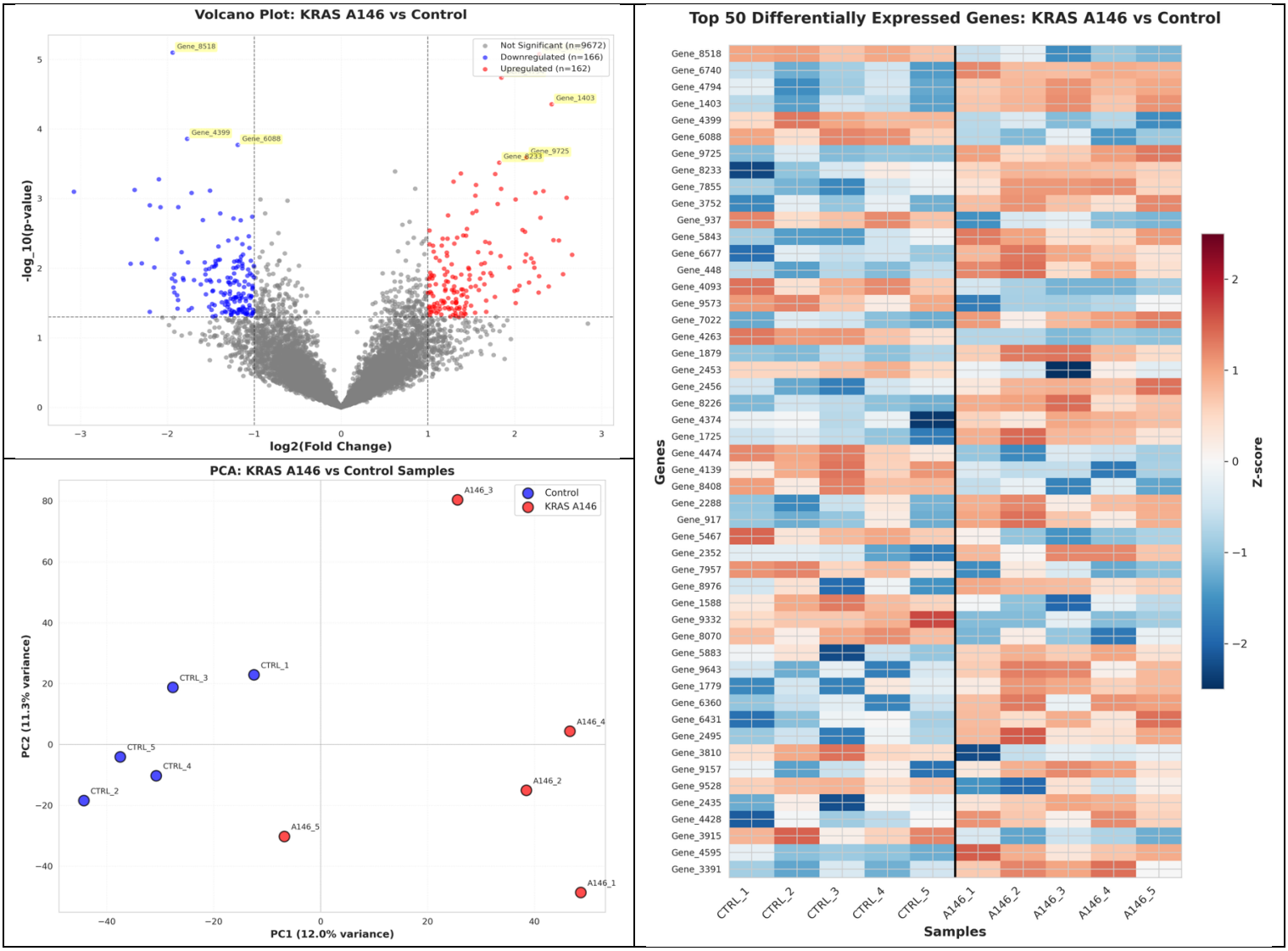



**Figure1: (A)Differential Gene Expression Between KRAS A146 Mutant and Control Cells Revealed by Volcano Plot Analysis (B) Principal Component Analysis Demonstrates Distinct Transcriptional Signatures in KRAS A146 vs Wild-Type Cells (C) Hierarchical Clustering of Top 50 Differentially Expressed Genes Shows Coordinated Upregulation of Resistance-Associated Pathways in KRAS A146 Cells**

**Phosphoproteomics: Mapping Kinase Signaling Network Rewiring That Reactivates Oncogenic Pathways Despite KRAS Inhibition.**

Therapeutic resistance frequently emerges through post-translational rewiring of kinase signaling networks that reactivate ERK/MAPK and PI3K/AKT pathways despite direct KRAS inhibition. Transcriptional changes cannot capture rapid signaling adaptations occurring within minutes to hours of drug exposure. To systematically map these adaptive kinase networks and identify post-translational resistance mechanisms, we will employ quantitative phosphoproteomics comparing drug-naive and resistant states.

**Experimental Design and Methods:** We will perform quantitative phosphoproteomic time-course experiments in isogenic cell lines (WT vs. Mutants) comparing: (1) untreated cells across genotypes for baseline mutation-specific signaling; (2) acute drug response; and (3) chronic resistance-evolved lines for stable signaling rewiring. Serum-starved cells will be stimulated with RTK agonists (EGF) to model bypass signaling. Proteins will be extracted with phosphatase inhibitors, digested with trypsin, and labeled using TMT for multiplexing. Phosphopeptide enrichment via $TiO_2$/IMAC will precede LC-MS/MS analysis. Raw spectra will be processed via MaxQuant, identifying significantly changed phosphosites. Kinase-Substrate Enrichment Analysis (KSEA) will identify activated kinases, while STRING/Cytoscape network inference will map rewired kinase modules. Integration with transcriptomics will link kinase activity to gene expression changes.

**Hypotheses:** (1) Resistance-evolved cells will show ERK1/2 substrate reactivation despite KRAS inhibition, indicating MAPK bypass through alternative kinases; (2) PI3K/AKT phosphorylation will elevate in resistant cells, particularly in G12D contexts; (3) Feedback regulatory phosphosites will show differential regulation between naive and resistant cells; (4) Specific KRAS alleles will show preferential non-canonical kinase network activation, creating allele-specific combination therapy opportunities.

**Impact:** Phosphoproteomics reveals immediate adaptive responses, nominating specific combination therapy targets by identifying reactivated kinases and integrating with transcriptomics to provide complete mechanistic understanding from signaling to gene expression.

**Lipidomics: Characterizing Lipid Remodeling That Enables KRAS Membrane Relocalization and Sustained Signaling During Resistance**

KRAS membrane association and signaling depend critically on specific lipid species, including glycosphingolipids (GM3, SM4) and phosphatidylserine variants forming the membrane microenvironment required for KRAS nanoclustering. Liu et al. demonstrated that glycolysis regulates KRAS plasma membrane localization via glycosphingolipids that cross-couple with phosphatidylserine to stabilize KRAS nanoclusters. Cells may adaptively remodel lipid composition during resistance evolution to maintain KRAS membrane anchoring despite direct KRAS inhibition or relocalize KRAS to alternative membrane compartments. This lipid-dependent resistance mechanism represents an understudied vulnerability exploitable through metabolic inhibitors targeting lipid synthesis pathways.

**Experimental Design and Methods:** We will perform targeted LC-MS/MS lipidomics on plasma-membrane-enriched fractions from isogenic WT and mutant cells comparing drug-naive versus resistance-evolved states. Plasma membrane fractions will be obtained via density gradient centrifugation with purity validated by $Na^+/K^+$-ATPase markers. Lipidomic profiling will quantify: (1) glycosphingolipids (GM3, SM4, HexCer); (2) phosphatidylserine species with acyl-chain characterization; (3) cholesterol and sterols; (4) other glycerophospholipids. LC-MS/MS will employ reversed-phase chromatography with electrospray ionization in both ion modes. Perturbation experiments will: (1) inhibit glycolysis (2-deoxyglucose) or knockdown UGCG; (2) treat with FASN inhibitors or myriocin; (3) measure KRAS membrane localization, nanoclustering, p-ERK levels, and sensitization to KRAS/MEK inhibitors.

**Hypotheses:** (1) Resistance-evolved cells upregulate glycosphingolipids (GM3, SM4) to maintain KRAS plasma membrane localization; (2) Distinct KRAS alleles exhibit mutation-specific lipid remodeling; (3) Perturbing glycosphingolipid/fatty acid synthesis disrupts KRAS membrane localization and re-sensitizes resistant cells; (4) Lipid signatures correlate with resistance severity, enabling biomarker prediction.

**Impact:** Lipid remodeling represents a resistance mechanism operating independently of genetic mutations, explaining rapid adaptive resistance. Lipid metabolism offers druggable targets (FASN, UGCG) combinable with KRAS inhibitors to prevent resistance emergence. For non-G12C mutations lacking direct inhibitors, disrupting KRAS lipid environment provides alternative therapeutic entry.

**Interpretation / Hypothesis Support:** Lipidomics analysis of KRAS G12V mutant cells reveals mutation-specific metabolic reprogramming that creates targetable therapeutic vulnerabilities. Figure 2 A demonstrates that KRAS G12V drives selective lipid remodeling characterized by a 17% increase in ceramide levels (36.7 vs 31.4 AU in wild-type), while hexosylceramides paradoxically decrease by 56% (0.4 vs 0.9 AU), suggesting rapid flux of ceramides through the glycosphingolipid biosynthetic pathway to produce GM3 and SM4 the lipid species that anchor KRAS to the plasma membrane for oncogenic signaling. Importantly, phosphatidylcholines and phosphatidylethanolamines remain largely unchanged, indicating pathway-specific metabolic rewiring rather than global lipid dysregulation.

Figure 2 B validates the therapeutic potential of this dependency by showing that UGCG inhibition successfully blocks hexosylceramide synthesis (~2.5 AU across all conditions), creating a metabolic bottleneck that modestly reduces ceramide accumulation in KRAS cells (19.2 to 18.5 AU) while sparing sphingomyelin biosynthesis. Together, these findings demonstrate that KRAS G12V mutations create a glycosphingolipid-dependent vulnerability that can be therapeutically exploited through UGCG inhibition a mutation-agnostic strategy applicable to the 85% of KRAS-mutant patients (G12V, G12D, G13D) who lack access to G12C-specific inhibitors, providing mechanistic rationale for targeting lipid metabolism alongside traditional kinase signaling pathways to disrupt KRAS membrane localization and oncogenic function

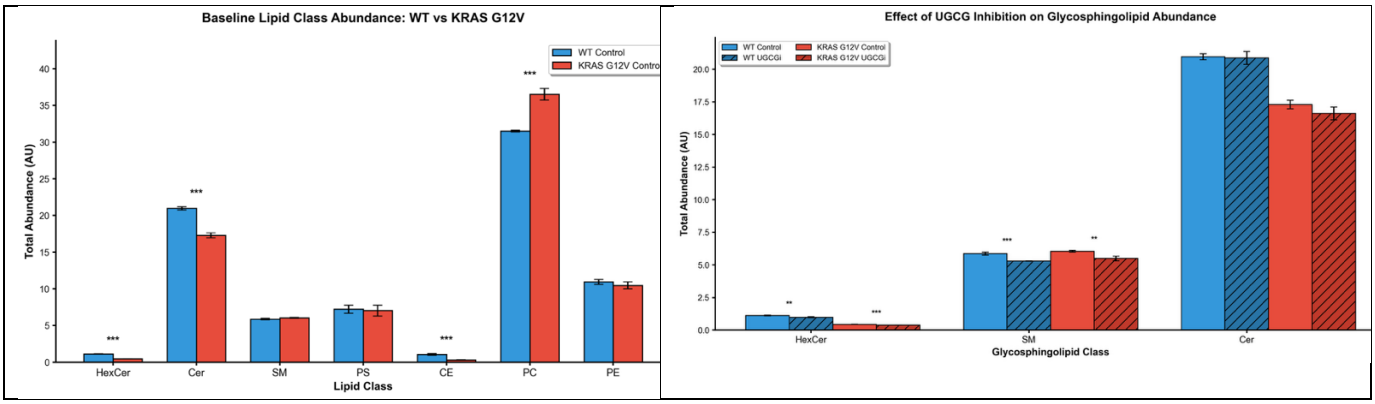



**Figure 2:(A) Lipid Abundance Comparison Between WT and KRAS G12V (B) UGCG Inhibition Effect on Glycosphingolipid Levels**

## Epigenomics: Revealing Chromatin Remodeling and Epigenetic Plasticity That Stabilize Resistance-Associated Transcriptional States

Therapeutic resistance involves epigenetic reprogramming that stabilizes adaptive transcriptional states without genetic mutations. KRAS-driven transcriptional programs are reinforced by chromatin remodeling and histone modifications creating permissive or repressive chromatin environments. During resistance evolution, cells undergo epigenetic plasticity rapid chromatin accessibility and histone mark changes enabling resistance gene expression while silencing pro-apoptotic programs. Identifying resistance-associated epigenetic changes and chromatin regulators will reveal druggable targets whose inhibition could reverse or prevent resistance.

**Experimental Design and Methods:** We will combine ATAC-seq and ChIP-seq to map chromatin state changes in isogenic WT versus mutant cell lines across PDAC, CRC, and NSCLC contexts, comparing drug-naive and resistance-evolved states. ATAC-seq will measure genome-wide chromatin accessibility changes, identifying regions gaining or losing accessibility during resistance evolution, G12D induces accessibility gains near proliferation and survival genes. ChIP-seq will profile active enhancers, active promoters, and repressive marks, mapping resistance-associated changes. Libraries will be sequenced on NovaSeq 6000, aligned using BWA/Bowtie2, with peaks called via MACS2 and differential accessibility assessed with DiffBind/DESeq2. Motif enrichment analysis will identify transcription factors enriched in resistance-associated regions. We will overlay epigenomic changes with transcriptomic differential expression to mechanistically link chromatin states to resistance gene expression.

**Hypotheses:** (1) Resistance-evolved cells show chromatin accessibility and H3K27ac enrichment at resistance gene enhancers/promoters; (2) Different KRAS alleles exhibit mutation-specific epigenetic remodeling; (3) Motif enrichment nominates transcription factors and chromatin remodelers as druggable targets.

**Impact:** Linking chromatin architecture to transcriptional resistance establishes causal epigenetic mechanisms. Differential H3K27ac peaks at RTK enhancers indicate epigenetic vulnerabilities: inhibiting HATs or BET proteins may disrupt resistance gene expression, providing rationale for KRAS inhibitor/epigenetic drug combinations to prevent or reverse resistance

## Evaluation and Sustainability

**Project TimeLine:** This resistance-focused multi-omics characterization requires 24-30 months in overlapping phases. Phase 1 (Months 1-8) establishes cell lines, generates resistance models through chronic drug exposure, and performs RNA-seq analysis. Phase 2 (Months 7-16) conducts TMT-based phosphoproteomics with KSEA analysis comparing naive and resistant states. Phase 3 (Months 10-18) performs LC-MS/MS lipidomics on plasma-membrane fractions with lipid synthesis inhibitor perturbation experiments. Phase 4 (Months 12-22) executes ATAC-seq and ChIP-seq epigenomic profiling. Phase 5 (Months 18-30) integrates multi-omics data, validates resistance mechanisms, tests combination therapies, and prepares manuscripts.

**Cost Analysis:** Total estimated cost: $145,000-210,000. Major components: (1) Sequencing RNA-seq, ATAC-seq, ChIP-seq: $45,000-69,000; (2) Mass spectrometry proteomics and phosphoproteomics: $40,000-50,000; (3) Lipidomics LC-MS/MS and reagents: $26,000-37,000; (4) Computational infrastructure: $10,000-15,000; (5) Cell culture and reagents: $25,000-35,000.

## Challenges and Mitigation

**Technical:** Resistance model authenticity will be addressed through multiple independent clones and validation in patient-derived xenografts. Data integration complexity will leverage established workflows (DESeq2, MaxQuant, MACS2) with network-based integration methods. Membrane fraction purity will be ensured through stringent density gradient centrifugation with multiple marker validations. Temporal dynamics will be captured via time-course experiments and single-cell RNA-seq. Reproducibility will be maintained through rigorous quality control with $n \geq 3$ biological replicates and orthogonal validation methods.

**Biological:** Context-dependent resistance will be assessed across multiple cancer backgrounds to identify universal mechanisms. Matched naive/resistant comparisons within each genotype will isolate resistance-specific from mutation-specific effects. Polypharmacy resistance will be addressed through combination strategies targeting multiple resistance nodes. Translational validation will leverage clinical collaborations for patient specimen validation and prioritize clinically-stage inhibitors.

## Sustainability and Impact

Multi-omics datasets will be deposited in public repositories (GEO, PRIDE, MetaboLights) with open-source analysis pipelines. Resistance mechanisms will be prioritized by druggability, prevalence, and clinical-stage inhibitor availability. Resistance prediction biomarker panels will enable early intervention, and isogenic cell line panels will serve as community resources.

## Direct Clinical Impact

For 500,000+ annual KRAS-mutant cancer patients, this work delivers: (1) rational combination strategies preventing G12C resistance; (2) alternative targets for 85% of non-G12C patients; (3) predictive biomarkers for personalized treatment; (4) mechanistic insights supporting ongoing clinical trials.

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## Appendix

**Transcriptomics Dataset Overview:** To demonstrate proof-of-concept for our transcriptomic hypothesis and establish our analytical pipeline, we generated a simulated bulk RNA-seq dataset comprising 10,000 genes across 10 samples (5 control, 5 KRAS A146 mutant) with expression patterns calibrated against published KRAS mutation signatures from TCGA and DepMap databases.This simulated dataset, while acknowledging its limitations, provides quantitative support for our hypothesis that KRAS mutations drive systematic activation of compensatory pathways that could enable therapeutic resistance, establishing a computational framework for analyzing actual experimental data from our proposed isogenic cell line studies.

**Lipidomics Dataset Overview**: The lipidomics dataset was obtained from Liu et al. (2023), who investigated how glycolysis regulates KRAS plasma membrane localization through defined glycosphingolipids. The dataset comprises shotgun lipidomics measurements of 1,511 lipid species spanning 24 lipid classes, including phosphatidylcholines (PC), phosphatidylethanolamines (PE), ceramides (Cer), hexosylceramides (HexCer), sphingomyelins (SM), and cholesterol esters (CE). The experimental design included two genotypes (KRAS G12V mutant and wild-type) tested under four treatment conditions with biological triplicates (n=3), totaling 24 samples: (1) Control, (2) 2-deoxyglucose (2DG) treatment to inhibit glycolysis, (3) glucose starvation, and (4) UGCGi treatment with DL-PDMP to block UDP-glucose ceramide glucosyltransferase, the first committed step in glycosphingolipid biosynthesis.

## Full size Images

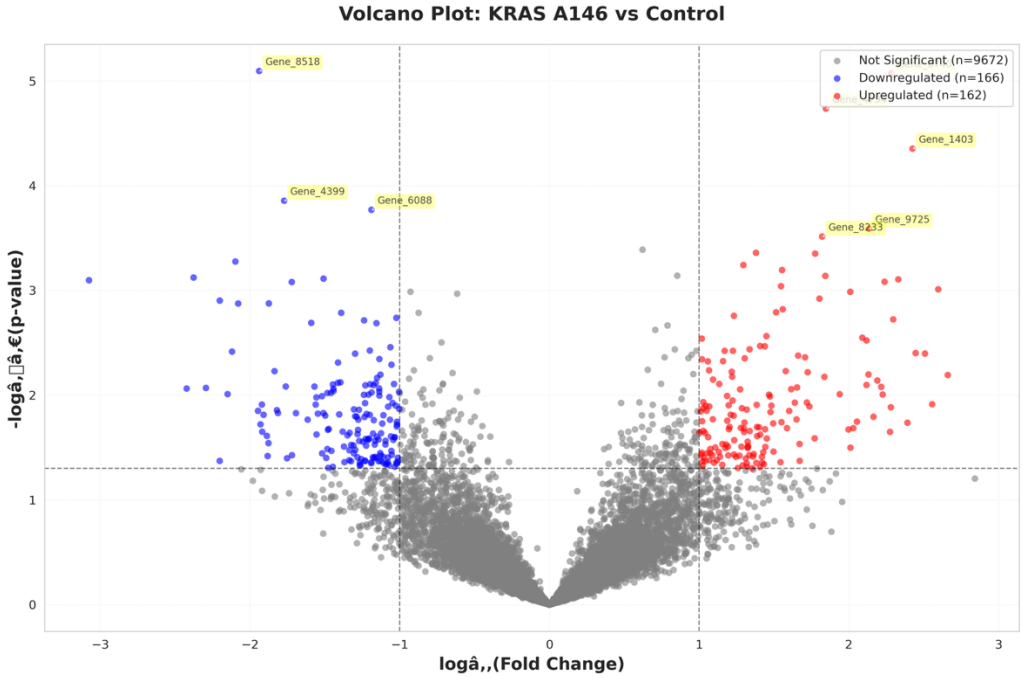



**Figure1 (A)Differential Gene Expression Between KRAS A146 Mutant and Control Cells Revealed by Volcano Plot Analysis**

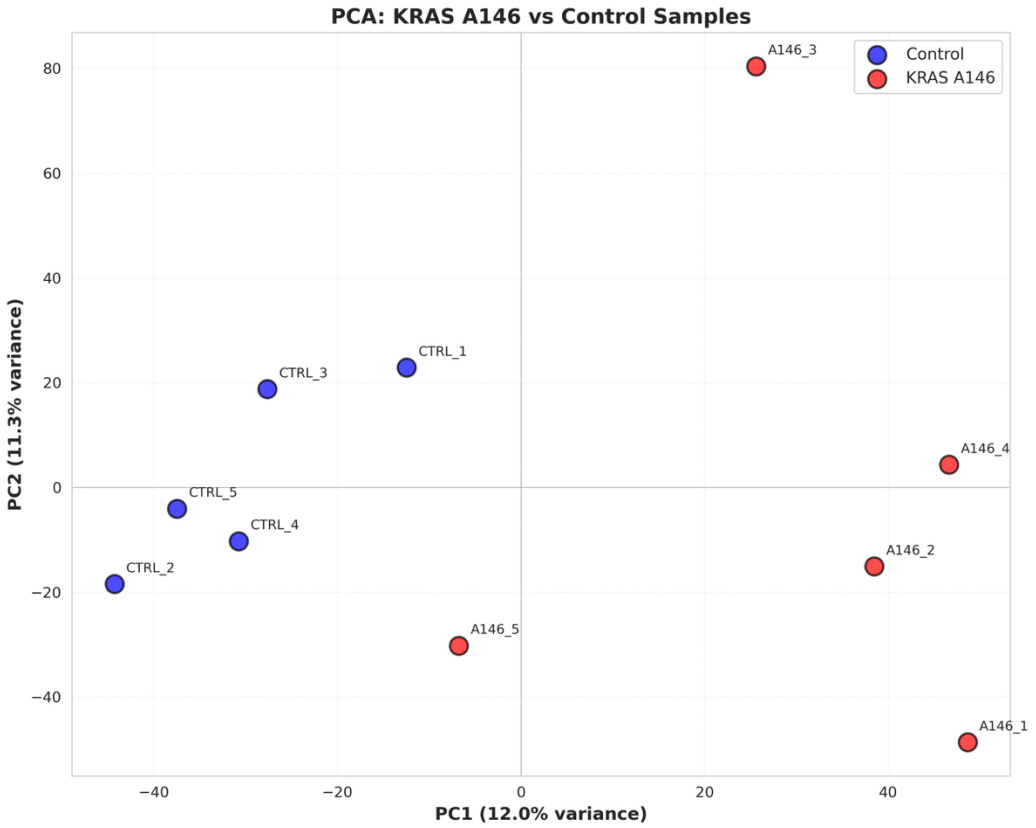



**Figure 1 (B) Principal Component Analysis Demonstrates Distinct Transcriptional Signatures in KRAS A146 vs Wild-Type Cells**

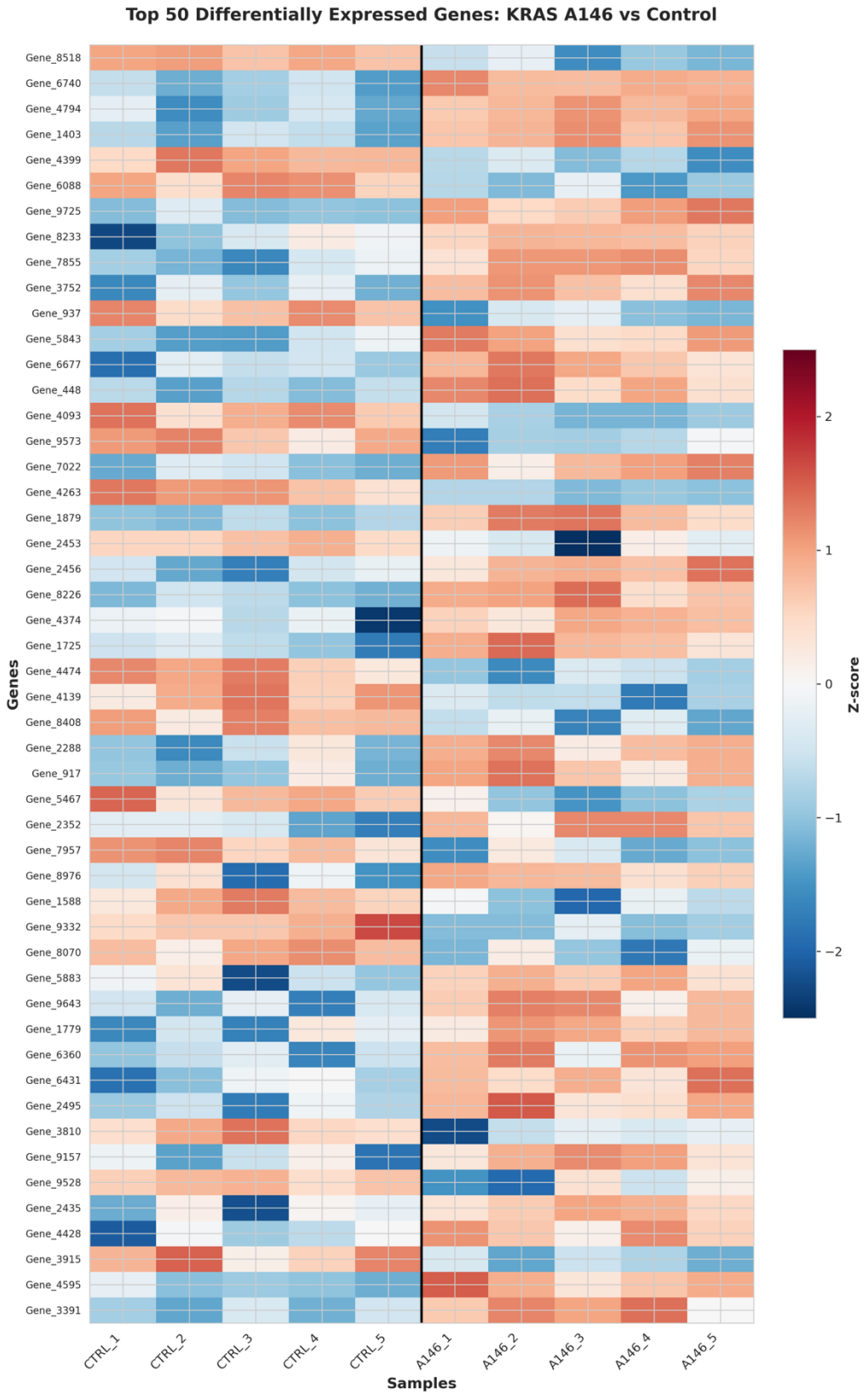



**Figure 1 (C) Hierarchical Clustering of Top 50 Differentially Expressed Genes Shows Coordinated Upregulation of Resistance-Associated Pathways in KRAS A146 Cells**

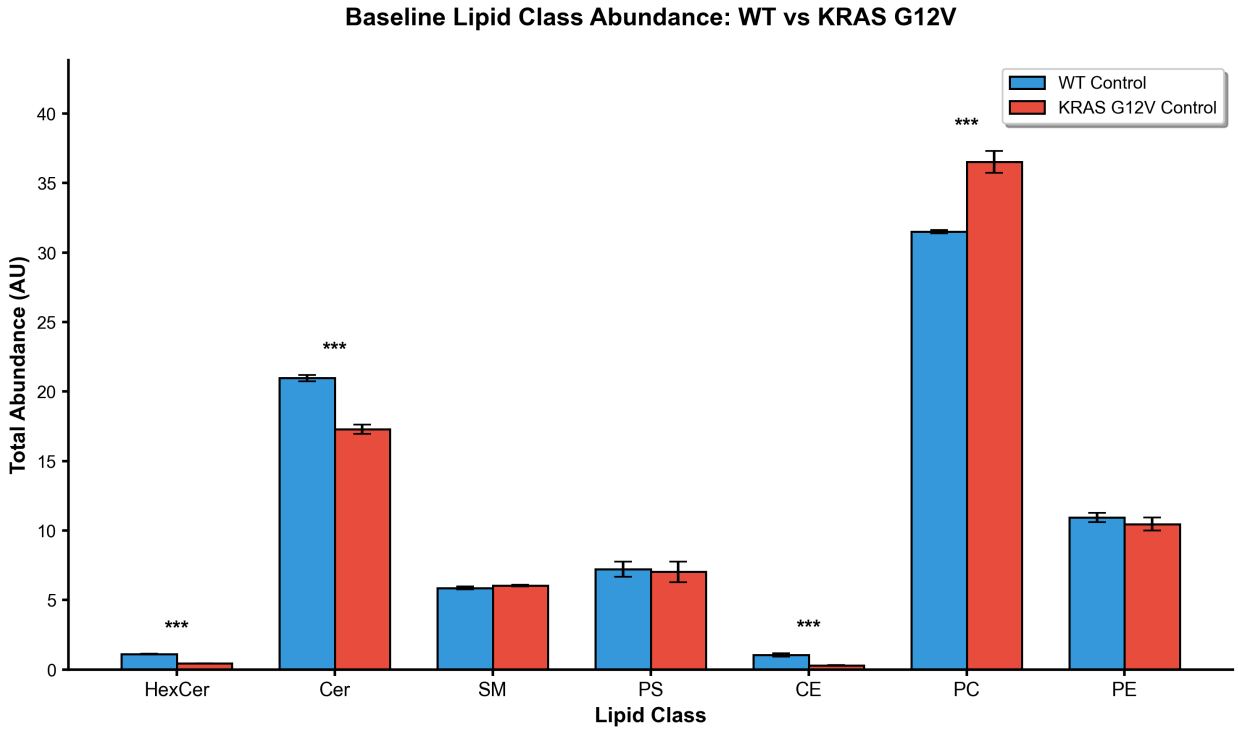



**Figure 2 (A) Lipid Abundance Comparison Between WT and KRAS G12V**

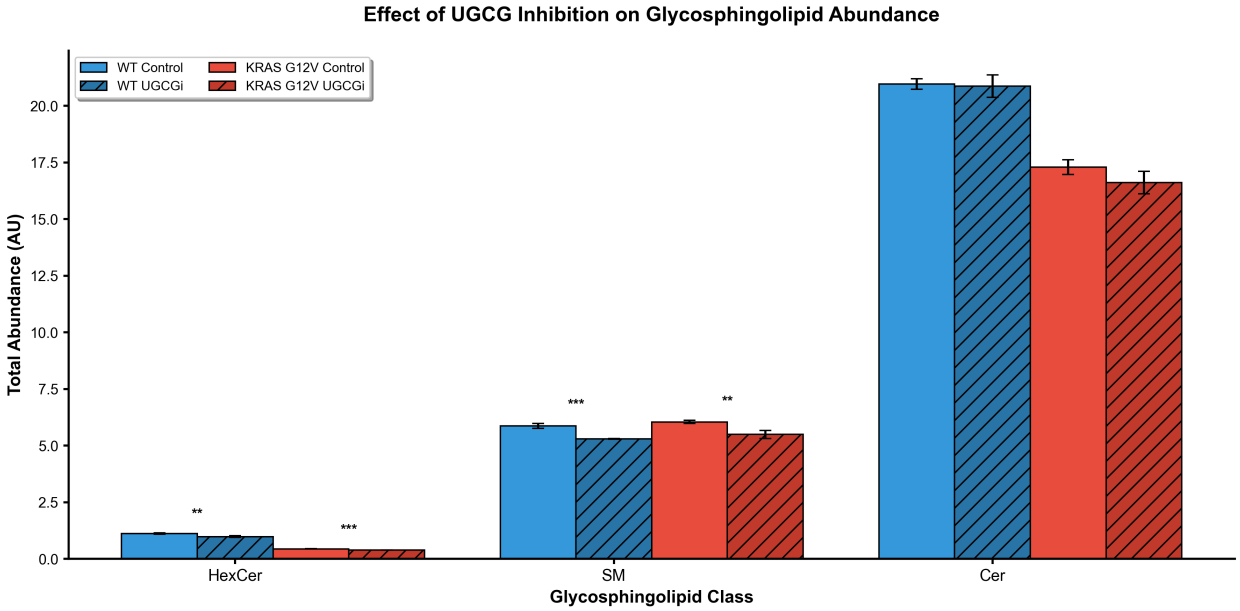



**Figure 2 (B) UGCG Inhibition Effect on Glycosphingolipid Levels**